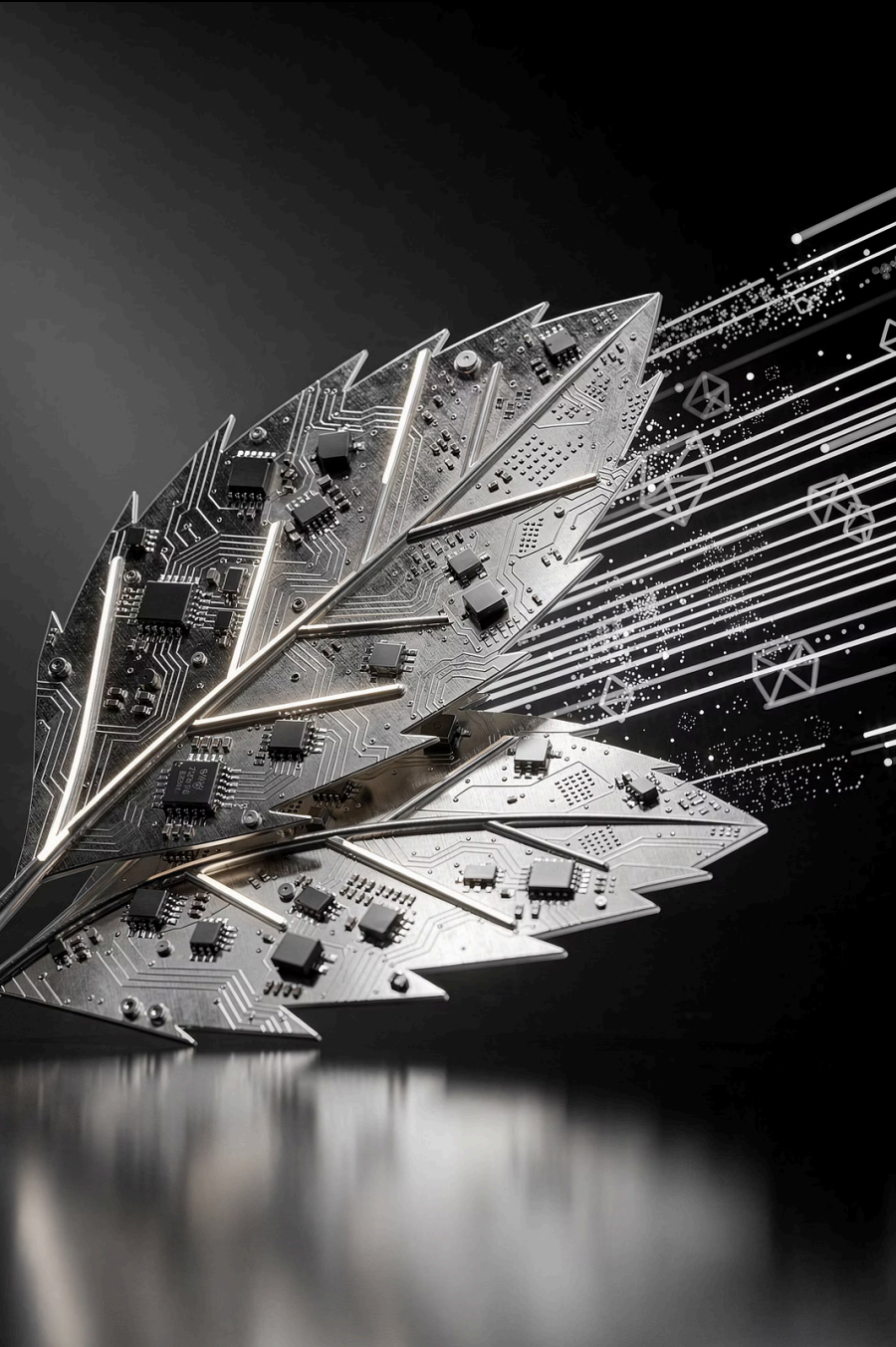




Green AI, When to Disengage, and Risks of Burnout

AI4Uni Training Session — Dr. Tymoteusz Miller

AI4UNI TRAINING SERIES



Why This Topic Matters

AI adoption in higher education is accelerating faster than most institutions can thoughtfully manage. Faculty and staff face mounting pressure to integrate AI tools into teaching, research, and administration — often without adequate guidance on sustainable use. The result is a perfect storm of competing demands that can undermine the very benefits AI promises to deliver.

Productivity Pressure

Do more with less — faster than ever before

Digital Overload

Too many tools, platforms, and inputs to manage

AI Dependency

Gradual erosion of independent critical thinking

Cognitive Fatigue

Mental exhaustion from constant context-switching

What is Green AI?

Green AI is a philosophy and practice of using artificial intelligence sustainably, responsibly, and intentionally. Rather than maximizing AI output at any cost, Green AI encourages users to ask: **Is this AI interaction necessary? Is it efficient? Does it serve a meaningful purpose?**

In academic contexts, Green AI means aligning your use of AI tools with both personal wellbeing and broader environmental responsibility — minimizing waste while maximizing genuine value.

Three Core Principles

→ Efficient Models

Choose the right tool for the task — not always the largest

→ Intentional Prompting

Think before you prompt — be deliberate and clear

→ Sustainable Balance

Integrate AI into human workflows, not the other way around



AI Has a Physical Cost

Behind every AI-generated response lies an enormous physical infrastructure that consumes real-world resources. Understanding this cost is the first step toward responsible, sustainable AI use in academic settings.



GPUs & Hardware

Powerful graphics processors run inference and training at massive scale, requiring rare materials and significant manufacturing energy.



Data Centers

Enormous facilities housing thousands of servers operate 24/7, distributed across the globe to serve real-time requests.



Cooling Systems

Preventing hardware from overheating demands industrial-scale cooling, often consuming millions of liters of water daily.



Energy & Water

A single large AI query can consume significantly more electricity than a standard web search, and water use rivals that of entire cities.

Everyday Academic AI Waste

Recognizing Wasteful Patterns

Academic AI waste is rarely dramatic. It accumulates quietly through small, habitual behaviors that feel productive but generate unnecessary computational and cognitive cost. Recognizing these patterns is the first step toward more intentional use.

Regenerating Outputs Repeatedly

Requesting multiple variations of the same content without clear criteria — hoping "the next one will be better" — wastes resources and rarely improves outcomes compared to editing a single good draft.

Unnecessary Prompts

Using AI for tasks that could be accomplished faster by the faculty member directly — simple calculations, brief emails, or information easily found in institutional systems.

Infinite Retry Behavior

Continuously re-prompting without refining the question, creating a loop of escalating frustration and resource consumption rather than pausing to think critically.

Green AI Practices for Academia

Adopting Green AI in your academic workflow doesn't require radical change — it requires intentional habits. These four strategies dramatically reduce waste while improving the quality of your AI interactions.



Better Prompts

Invest time crafting clear, specific, well-structured prompts. One good prompt outperforms ten vague attempts — saving time, energy, and compute.



Smaller Models

Match the model to the task. Lightweight AI tools are sufficient for most academic drafting and summarization tasks — reserve large models for complex challenges.



Reuse Templates

Build and maintain a personal library of effective prompts and AI-assisted templates. Reusing proven approaches eliminates repetitive generation cycles.



Human-First Workflows

Design your process so that human judgment leads and AI assists. Start with your own thinking, then use AI to refine — not the other way around.



AI and Cognitive Load

AI tools are designed to reduce manual effort — but they can simultaneously increase **mental fragmentation** when integrated without boundaries. The cognitive cost of managing AI-assisted workflows often goes unrecognized until it becomes a serious problem.

1

Notification Overload

AI platforms generate constant status updates, suggestions, and alerts that interrupt sustained academic thinking and fragment concentration across tasks.

2

Context Switching

Jumping between AI interfaces, course platforms, email, and documents forces the brain to repeatedly reload context — a cognitively expensive process that drains working memory.

3

Prompt Fatigue

The mental effort of continuously formulating, evaluating, and revising prompts accumulates throughout the day, creating a unique form of intellectual exhaustion distinct from traditional work fatigue.

What is AI Burnout?

AI burnout is an emerging phenomenon distinct from general digital burnout. It arises specifically from **chronic, high-frequency engagement with AI tools** in high-stakes professional contexts — precisely the environment many university faculty occupy.

Unlike traditional burnout, AI burnout can be insidious: it often begins while productivity metrics appear strong, making it easy to dismiss early warning signs until the individual is significantly depleted.

⚠️ AI burnout can develop even when — and sometimes because — AI appears to be working well.

Exhaustion

Persistent mental and emotional depletion that rest alone doesn't resolve

Reduced Creativity

Difficulty generating original ideas; defaulting to AI output rather than personal insight

Cynicism

Growing skepticism or hostility toward AI tools, colleagues, or the institution's digital agenda

Emotional Fatigue

Declining sense of professional meaning and purpose in AI-augmented work

Academic Environment Risk Factors

Higher education creates a uniquely high-risk environment for AI burnout. The confluence of longstanding academic pressures with the rapid adoption of AI tools amplifies vulnerability in ways other professional settings do not.

1

Publish-or-Perish Pressure

Relentless research output expectations drive faculty to use AI as a speed tool — increasing volume without reducing underlying stress

2

Administrative Overload

AI is deployed to manage growing administrative burdens, but each new tool adds complexity, training demands, and decision overhead

3

Constant Adaptation

Rapid AI evolution requires continuous skill updates — keeping pace becomes exhausting when there is no stable end point in sight



The Productivity Trap

When efficiency tools increase expectations rather than reduce workload, something has gone wrong.

The Promise

AI is adopted with the expectation that it will free up time, reduce repetitive work, and allow faculty to focus on higher-value intellectual activities — research, mentoring, creative scholarship.

The Reality

In practice, efficiency gains are frequently absorbed by escalating demands. Departments expect more output. Courses require more frequent updates. Administrative workflows multiply. The "time saved" evaporates — while the cognitive burden of managing AI tools is added on top.

 Efficiency gains that increase output expectations without reducing workload are not a benefit — they are a trap. Sustainable AI use requires institutional boundaries, not just individual willpower.

When AI Stops Helping

AI tools have a point of diminishing return. Recognizing when you have crossed that threshold is a critical professional skill. These warning signs indicate that your current AI engagement is costing more than it contributes.



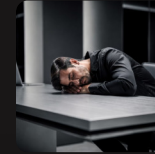
Aimless Prompting

Opening an AI interface without a clear goal — browsing prompts hoping for inspiration rather than solving a defined problem. This signals cognitive avoidance, not productive use.



Overreliance

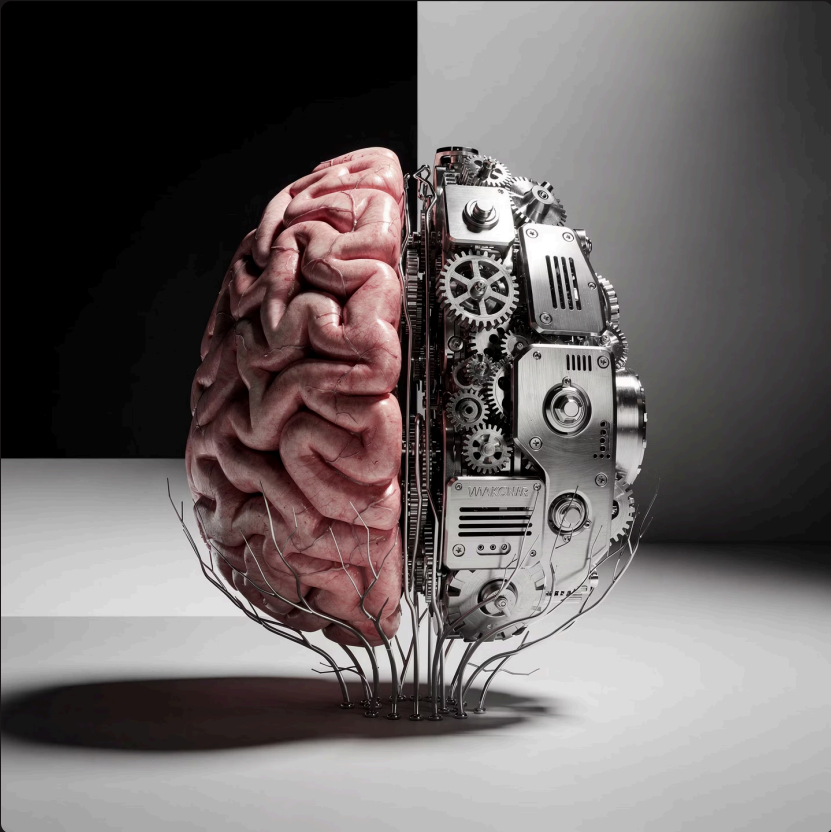
Feeling unable to begin tasks — writing, analysis, planning — without AI assistance first. When AI shifts from a tool to a prerequisite, intellectual independence is at risk.



Mental Exhaustion

Feeling more drained after AI-assisted work sessions than before them. If AI interaction is consuming rather than restoring cognitive energy, the workflow needs to change.

Cognitive Dependency Risks



Heavy reliance on AI tools — even high-quality ones — can gradually reshape cognitive habits in ways that weaken long-term professional capability. These risks are not hypothetical; they are observable in early research on AI-augmented knowledge work.

Reduced Analytical Thinking

Consistently delegating analysis to AI weakens the practitioner's own ability to reason through complex problems independently over time.

Automation Bias

A well-documented tendency to trust AI outputs over one's own judgment — even when the AI is clearly wrong — because the output appears authoritative and effortless.

Passive Acceptance

Accepting AI-generated content without critical engagement, embedding errors, mediocrity, or misaligned perspectives into scholarly work.

AI Hallucinations & Mental Fatigue

Every AI interaction carries a verification cost — and that cost accumulates.

AI language models can generate plausible-sounding but factually incorrect content — citations that don't exist, statistics that are fabricated, logical errors embedded in fluent prose. In academic work, undetected hallucinations carry serious professional consequences.

The hidden burden is the **cognitive labor of continuous verification**. Responsible AI use in academia requires actively and critically evaluating every AI output — a task that is intellectually demanding and easy to let slip when fatigued. Over time, the pressure to verify everything AI generates becomes a significant additional workload, not a reduction of it.

The Verification Cycle

- 1 AI generates output
- 2 Faculty must verify accuracy
- 3 Fatigue reduces verification rigor
- 4 Errors enter scholarly work

Healthy AI Boundaries

Setting intentional boundaries around AI use is not a rejection of technology — it is a professional strategy for sustainable, high-quality academic work. The goal is to remain the author of your work in the fullest sense.



Deep Work Blocks

Schedule dedicated time for focused intellectual work with all AI tools and notifications off. Guard these sessions as non-negotiable. Deep thinking requires uninterrupted time.



Scheduled AI Use

Rather than reaching for AI reflexively throughout the day, designate specific time slots for AI-assisted tasks. This prevents AI from fragmenting your cognitive workflow.



Protected Thinking Time

Before prompting, spend a few minutes thinking independently. Formulate your own perspective first. Use AI to refine and pressure-test your thinking — not to replace it.

Human Skills AI Cannot Replace Well

The most durable professional investment faculty can make is strengthening the capabilities that AI cannot replicate. These are not soft skills — they are the core of what makes academic work meaningful, trustworthy, and transformative.



Empathy

Understanding students' personal contexts, struggles, and motivations in ways that build genuine relationships and support real learning.



Ethical Judgment

Navigating the complex moral dimensions of research, pedagogy, and institutional life with nuance, wisdom, and personal accountability.



Mentoring

Shaping the intellectual identity and professional trajectory of students through sustained personal engagement that no AI interaction can substitute.



Scientific Intuition

The pattern recognition, disciplinary instinct, and creative hypothesis generation that emerges from years of immersive scholarly experience.

Institutional Responsibility

Sustainable AI use in higher education cannot rest solely on individual faculty choices. Universities bear a structural responsibility to create the conditions under which healthy, ethical, and effective AI integration can occur across the institution.

Create AI Policies

Develop clear, evidence-based institutional guidelines governing AI use in teaching, research, assessment, and administration — providing faculty with frameworks rather than leaving them to navigate alone.

1

2

3

Encourage Sustainable Usage

Actively model and reward intentional, sustainable AI practices — rather than implicitly incentivizing maximum AI adoption as a proxy for innovation or efficiency.

Promote Digital Wellbeing

Integrate digital wellbeing into faculty development programs.
Recognize and address AI burnout as a legitimate occupational health concern, not a personal failing.

i Institutions that invest in AI governance and faculty wellbeing today will build more resilient, trusted, and effective academic communities tomorrow.

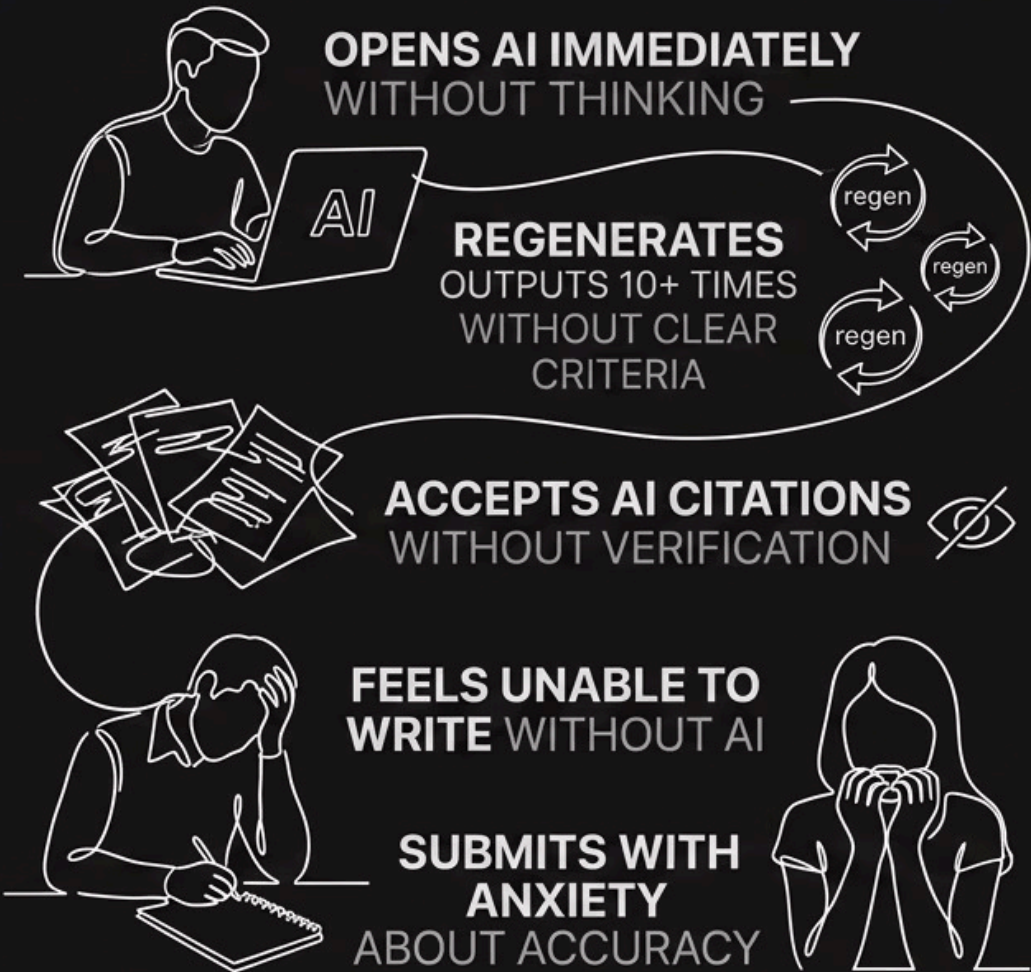
Healthy vs. Unhealthy AI Usage

Understanding the difference between productive and problematic AI integration is easier with concrete scenarios. The following comparisons reflect real patterns observed in academic AI use.

Healthy AI Usage



Unhealthy AI Usage



The distinction between these workflows is not about how much AI is used — it is about **who is leading the process**. In healthy usage, the faculty member's judgment drives every stage. In unhealthy usage, AI gradually assumes that role.

Practical AI4Uni Recommendations

These evidence-informed recommendations form a practical starting point for every faculty member navigating AI integration. Apply them as a personal checklist and revisit regularly as your AI use evolves.

1

Use AI Intentionally

Define a clear purpose before every AI interaction. Ask: *What specific problem am I solving, and is AI the right tool for it?*
Purposeless prompting wastes resources and erodes focus.

2

Verify All Outputs

Never publish, teach, or submit AI-generated content without independent verification. Cross-check facts, citations, and logical claims against authoritative sources every time.

3

Reduce Digital Noise

Consolidate your AI tools to the essential few. Turn off non-critical notifications during deep work. A simpler digital environment directly supports clearer, higher-quality thinking.

4

Protect Your Thinking

Maintain regular AI-free periods in your weekly schedule. These protected intervals preserve cognitive independence and sustain the creative capacity that defines excellent scholarship.

Final Reflection

The goal is not maximum AI usage. The goal is **sustainable and meaningful AI usage** — in service of deeper thinking, better teaching, and more humane scholarship.

What We Are Moving Away From

- AI as a reflexive first response to every task
- Efficiency as the primary measure of academic value
- Digital tools that fragment attention and erode judgment
- Institutional cultures that ignore cognitive cost

What We Are Moving Toward

- AI as a deliberate, bounded tool within human-led workflows
- Sustainable practices that protect creativity and wellbeing
- Institutions that model responsible digital citizenship
- Scholarship that remains deeply, irreducibly human





Discussion & Q&A

Thank you for your engagement with today's session. The following questions are offered as starting points for reflection and discussion — bring your own experiences, challenges, and perspectives to the conversation.

Green AI

Where in your current workflow do you see the most AI waste? What one change could make the biggest difference?

Burnout & Boundaries

Have you noticed any of the warning signs of AI burnout in your own work? What boundaries have you already found helpful?

Institutional Role

What would you want your institution to do differently to support sustainable AI integration across the faculty?